

Vishwanath Maddula

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EDUCATION

Masters in Computer Science - University of Illinois at Springfield | GPA: 4.0/4.0

Aug '23 - Dec '24

Coursework: Advanced Java Programming, Advanced Database Concepts, Computer Networking Principles, Digital Forensics, Computer Security, Security Testing Essentials, Graduate Research Seminar.

Bachelor's in Computer Science and Engineering - Amrita Vishwa Vidyapeetham | GPA: 9.2/10

Jul '19 - Jun '23

Relevant Coursework: Object-Oriented Programming, Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks, Distributed Systems, Theory of Computation, Data Analytics, Machine Learning.

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL, JavaScript

Frameworks & Libraries: Spring Boot, Spring Cloud, Flask, ReactJS, Airflow, NLTK

APIs & Architecture: RESTful APIs, Microservices, Event-Driven Systems, OAuth Integrations

Databases: PostgreSQL, MySQL, Elasticsearch

Cloud & DevOps: GCP, Docker, Git, GitHub Actions, Maven, CI/CD

Testing & Tools: JUnit, Pytest, Postman

Data Engineering: Kafka, Pub/Sub, ETL/ELT Pipelines, Data Quality (SODA)

Web Technologies: HTML, CSS, JSON, XML

Analytics & Reporting: Power BI, MS Office Suite

OS & Cybersecurity: Linux, Windows, TCP/IP, OSI Model, OWASP best practices

WORK EXPERIENCE

Software Engineer - Data Analytics Platform, FoodSupply AI

Mar '25 - Present

- Engineered reusable, platform-specific data ingestion connectors using **Java, Python, and GCP**, reliably processing over **3GB of daily marketing data** from external APIs.
- Developed stream-processing logic using **Spring Cloud Stream** and **Google Pub/Sub**, standardizing **100K+ records/day** into a canonical schema required for analytical systems.
- Managed a multi-tier data storage architecture utilizing **GCP BigQuery** for analytics, **GCS** for data lake storage, **Elasticsearch** for search and logging, and **Cloud SQL** for transactional needs.
- Wrote, reviewed, and optimized complex **SQL** queries across various data stores, resulting in a **~30% improvement** in aggregation query performance.
- Automated data workflows using **Cloud Composer**, maintained **Spring Boot REST APIs**, and resolved production issues to ensure platform stability and data consistency.

Software Engineer - Data Platform, Honeywell Aerospace

Jan '23 - Jul '23

- Migrated file-based runway data retrieval processes to **PostgreSQL**, improving search accuracy and reducing lookup time.
- Developed a data-driven web application using **Java, Spring Boot**, and REST APIs, reducing manual data search effort by **8-10 hours per week** for engineering teams.
- Built scalable microservices using **Spring Boot, Spring MVC, Spring Cloud, Eureka, and OpenFeign** to streamline data access across systems.
- Contributed to the **design and optimization of relational data models** using **database normalization techniques**.
- Automated builds and deployments using **Maven** and integrated them into CI/CD pipelines for smoother releases.
- Ensured production-grade reliability through **JUnit tests (95% coverage)** and API validation using Postman.
- Developed **Python-based** pattern-matching algorithms using **NLTK** to extract from **25+ years** of archived aerospace data.

PROJECTS

Construction Company Management System | [Link](#)

Skills: Use Case Analysis, Database Design, Database Normalization, System Design, LLD, Java Software Development

- Developed** construction company management software to model the operational needs of a company, including project tracking, employee management, and resource allocation.
- Designed** an **ER diagram** with **15 entities** and used **Java, Java Swings**, and **PostgreSQL** for developing the system.

YouTube Data ELT Pipeline using Airflow | [Link](#)

Skills: Python, SQL, Airflow, Docker, PostgreSQL, YouTube API, SODA Core, Pytest, GitHub Actions

- Built an **ELT pipeline using Airflow and Docker** to extract and process YouTube data into PostgreSQL.
- Implemented **data quality checks** with SODA and **unit tests** using **pytest**.
- Set up **GitHub Actions CI/CD** for automated Docker image builds and Airflow deployment.

Task Failure Prediction in Cloud Data Centers | [Link](#)

Skills: Exploratory Data Analysis, Machine Learning, Deep Learning, Backend Development

- Built an ensemble voting classifier model** (SVM, Decision Trees, Random Forest) for early failure prediction in data center tasks, training on the **Google Cluster Trace (GCT) 2019** dataset.
- Achieved 92.5% prediction accuracy**, providing actionable insights for task rescheduling.